

Module 3 Reflection Journal

In this lab exercise, we learned about the three main types of machine learning: Supervised, Unsupervised, and Reinforcement learning. Supervised learning is the most interesting because it can classify new data using labeled examples. This is particularly useful in the field of cybersecurity, where a model can learn from previously labeled network activities and then identify whether new activities are normal or potentially malicious.

The most important part of the machine learning process is learning how to separate data into a training set and a testing set. The training set is used to teach the model, while the test set is used to evaluate the model's performance using data it hasn't seen before. The reason why this separation is important is that if you only test the model on the data it was trained on, it won't reliably show how well it can handle new data.

The hardest part of this lab is interpreting the results of the model. Accuracy can give a general score, but the classification report and confusion matrix show more detailed information about the correct predictions and mistakes. We learned that when evaluating a machine learning model, it's important to consider many different metrics instead of relying on just one accuracy score.

For my experiment, I chose color intensity, the optical-density ratio of Cheesewines, and proline as the three main input features. My Logistic Regression model achieved 100% accuracy on the test set, while the original Logistic Regression model had 88.9%. This shows that the selected features contain strong information to distinguish the three types of wine. However, getting perfect results on just one test split doesn't prove that the model will perform perfectly on all new data. To determine if the model's performance is stable, I'll use cross-validation to test it with different data splits.

I could apply these concepts to cybersecurity by developing a model that detects suspicious network activity. The model could use information such as IP addresses, ports, network protocols, connection duration, packet counts, and unsuccessful login attempts. A supervised classification

model could learn from network events labeled as normal or malicious and then help security analysts identify possible attacks.

Overall, this lab has helped me understand the entire process of data preparation, feature selection, data sharing, training models, performance evaluation, and interpretation of results. In the future, I would like to explore how feature scaling and cross-validation affect model reliability, as well as how evaluation metrics are best suited for cybersecurity data sets, where risky processes are often far lower than normal processes.